

Expectation vs Reality in Users' Willingness to Delegate to Digital Assistants

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ABSTRACT

We report results of a survey with 2500 US-based users of popular digital assistants (DAs) to understand and prioritize the factors that drive consumer adoption. Using structural equation modelling, we investigated the relationship between respondents' behavioural intentions to delegate tasks to their DAs and two predictor layers: DA (users' attitudes and familiarity with them) and task factors (need for control/transparency, subjectivity, risk, self-efficacy, and frequency), mediated by a values layer (trust, perceived ease of use and usefulness). Perceived usefulness and trust were strong direct predictors of willingness to delegate. Both DA-related factors had indirect effects, with a surprising negative influence of familiarity on both trust and usefulness. Our novel findings of task-related effects on the value factors may explain this disparity. These results imply a mismatch between users' expectations for DAs and their actual experiences. We interpret these findings in light of related work and derive implications for practitioners.

CCS CONCEPTS

• **Human-centered computing** → **HCI theory, concepts and models.**

KEYWORDS

digital assistant, willingness to delegate, trust, technology adoption, expectations, SEM

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*Work done while the author was at Google

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1 INTRODUCTION

The number of digital voice assistants (DAs), such as Apple's Siri or Google Assistant, in use globally is forecast to exceed eight billion—roughly equivalent to the global population—by 2024 [44]. DAs, powered by artificial intelligence (AI), intend to help people accomplish various tasks and come in a variety of devices, ranging from mobile phones and smart speakers to wearable devices and car displays. Despite their ubiquity and growing capabilities, DAs are far from being used to their full potential. Playing music, running basic search commands, and controlling lights persist as the main uses of DAs, while their users remain reluctant to explore more advanced tasks [3, 20].

Recent qualitative studies and systematic literature reviews [8, 34, 43] provide a comprehensive overview of the factors contributing to users' engagement with a range of AI agents. However, the relative importance of these factors in individuals' motivation or reluctance to use DAs for particular tasks is not well understood. A quantitative investigation is necessary to identify the main drivers of these behavioural intentions and prioritize areas for DA improvement.

We address this need in our present work. Our contributions are three-fold: 1) We devise a unified model of key factors predictive of users' willingness to delegate tasks to DAs based on a comprehensive literature review. Our model combines an extensive list of predictor factors describing users' perceptions of both DAs and tasks, as well as perceived value of DAs derived from these predictors (Section 3). 2) We report how we validated this model using psychometric techniques [37] and structural equation modelling (SEM) [19, 25]. The resulting model demonstrates the relative importance of and relationship between its comprising factors (Section 4). 3) Based on the validated model, we outline implications to guide practitioners to deliver more helpful technology to DA users (Section 5).

2 RELATED WORK

Automation research refers to 'delegation' of a task to an agent by the user [9]. Lubars and Tan [31] presented a framework of task delegability characterizing human preferences for AI automation across tasks. Castelo et al. [10] studied users' 'algorithm aversion' depending on the type of task that they face. Bouwer [7] devised a taxonomy of human emotional states which drive the motivation to delegate to AIs. This taxonomy is useful to identify which tasks

people are willing to entrust based on their emotional appraisals of those tasks and their outcomes.

‘Delegation’ implies relying on a surrogate to complete a task. A large body of quantitative work has explored interpersonal delegation in contexts ranging from management decisions to shopping [e.g. 1, 29, 30]. SEM is a valuable procedure for this research because it allows researchers to assess hypothetical unobserved constructs (e.g. willingness to delegate) and study possible causal relationships between such factors with path analysis [25]. For instance, Aggarwal et al. used a path model to analyse delegation of purchase decisions [1]. Similar approaches have been used to study delegation to AI agents, such as customers’ willingness to accept AI device use and their behaviours in service encounters [18], or effects of personalization and familiarity on trust and adoption of recommendation agents [26]. Some studies also draw on concepts from technology adoption models such as TAM [15], for instance by modelling consumers’ attitude and intention to use smartphone chatbots for shopping [22] or customers’ relations with AI-powered chatbots for hospitality and tourism [40]. Most of this work has focused on custom AI tools for specific contexts. General-purpose DAs are the subject of valuable qualitative research [e.g. 14, 32, 45]. However, SEM-based approaches to identify predictors of willingness to delegate, such as those cited above, are lacking for DAs.

We addressed this gap by defining and testing a model of willingness to delegate to DAs that took into account different types of tasks as well as factors driving technology adoption. Based on the discussed work, we focused our study on areas where automation via DAs has the most potential value. We concentrated on tasks that users are more likely to consider for delegation and selected factors which can be influenced by technology creators.

3 MODEL DEVELOPMENT

Our goal was to understand what drives users’ willingness to delegate tasks to digital assistants. We drew inspiration from the Technology Acceptance (TAM) [15] and Value-based Adoption (VAM) [24] Models. We combined them with exogenous factors describing users’ perceptions of the DAs and task characteristics. Therefore, our model included four main areas of focus: users’ perception of the DAs, users’ perceptions of the tasks they face, users’ perceived value of DAs, and willingness to delegate. Below, we explain the factors comprising these areas and describe our hypotheses about their relationships.

3.1 Users’ perceptions of DAs

Users’ perceptions of DAs describe users’ appraisals and interaction experiences with this technology. Prior research has identified that users’ attitudes towards DAs [22] and users’ familiarity with DAs [26] play an important role in defining users’ behavioural intentions. Therefore, we included these two factors as exogenous variables in our model. Their definitions are provided in Table 1.

3.2 Users’ perceptions of tasks

Users’ perceptions of tasks describe task characteristics regardless of whether users use DAs to complete them or not. These factors contribute to people’s delegation decisions to surrogates in general [1], and AI-driven agents in particular [10, 31]. Table 1 includes

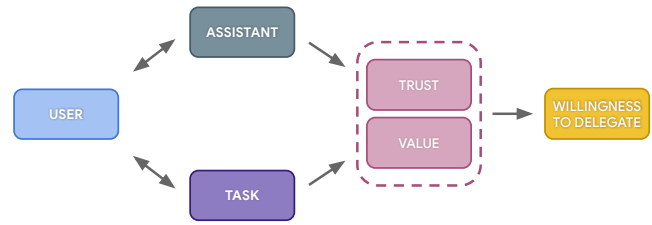


Figure 1: High-Level Hypothetical Model.

definitions of the six considered factors: users’ need for control and transparency, task riskiness, perceived task subjectivity, users’ self-efficacy, task complexity, and task frequency.

3.3 Perceived value of DAs

The values layer in our model is inspired by the VAM [24], where value defines an overall estimation of the choice and acts as a mediator between the exogenous factors and individual decision-making. Rather than considering one holistic factor, we included two factors from the TAM [15], perceived usefulness and perceived ease of use, as well as a factor describing trust in DAs, based on prior research [22, 40]. We provide the definitions of these factors in Table 1.

3.4 Behavioural intentions

We frame behavioural intentions as **users’ willingness to delegate** a task to DAs. Although it does not capture the behaviour per se, behavioural intention can be a strong predictor of actual behaviour [2].

3.5 Hypotheses

To evaluate our model, we formed a set of hypotheses about how various factors relate to each other (Figure 1). Based on prior research, we hypothesized that users’ perceptions of DAs have a positive effect on trust and value factors. Perceived task characteristics were also expected to have an influence on these factors: task complexity and frequency a positive influence, the other factors a negative one. Finally, we posited a positive effect from trust and value factors of DAs to users’ willingness to delegate. The detailed hypotheses and supporting references are described in Table 1.

3.6 Tasks

To achieve clarity and consistency in measuring the outlined factors through a survey, we used vignettes describing 8 different task scenarios to anchor participants’ responses. To make sure that we evenly covered all potential opportunities for user interaction with DAs, we selected 2 tasks representing each of the 4 classes of automation [38]: acquisition, analysis, decision, and action. The 8 tasks are listed in Table 2. We treated the tasks as control variables, so we included them as exogenous categorical variables leading to task-related factors in the model. This allowed us to make sure that the paths between main factors reflect their relationships independent of the task.

Table 1: Factors, definitions and hypothesized paths.

Category	Factor and Definition	Hypothesized path
Users' perceptions of DAs	Users' attitude towards DAs: users' negative or positive feelings towards DAs [22]	+ Trust [27]* + Perceived Ease of Use [12, 46]* + Perceived Usefulness [5, 12]*
	Users' familiarity with DAs: users' understanding of digital assistants based on their previous interactions [26]	+ Trust [26, 43] + Perceived Ease of Use [13, 43] + Perceived Usefulness [13, 43]
Users' perceptions of tasks	Users' need for control and transparency: users' desire to stay aware and in control of the decision-making process while completing the task [1, 31]	- Trust [1, 31]
	Task riskiness: the probability of harm or loss based on the degree of importance and uncertainty associated with the task [31]	+ Users' need for control and transparency [31]
	Perceived task subjectivity: the degree to which the outcome of the task is open to interpretation and depends on personal opinion or intuition [10, 31]	- Trust [10] - Perceived Usefulness [10]
	Users' self-efficacy: the level of users' confidence in their ability to complete the task without help [31]	- Perceived Ease of Use - Perceived Usefulness [1, 31]
	Task complexity: the degree of effort and skills required to complete the task [31, 34]	- Trust + Perceived Usefulness [1, 31]
	Task frequency: the rate at which the users face the task [16]	+ Perceived Usefulness [16]
Perceived value of DAs	Trust in DAs: users' trust in a DA to complete a task, with trust being defined as the willingness to take a risk based on the expectation of a benefit [35]	+ Willingness to delegate [1, 6, 22, 40]
	Perceived ease of use of DAs: the degree to which users believe that using a digital assistant to complete a task would be free of effort [15]	+ Perceived usefulness [6] + Willingness to delegate [4, 18, 40]
	Perceived Usefulness of DAs: the degree to which users believe that using a digital assistant would enhance their performance on a task [15]	+ Willingness to delegate [6, 18]
Behavioural Intentions	Willingness to delegate: willingness to use DAs to complete a task	

* In prior work, attitudes served as mediator between exogenous variables and behavioral intentions. However, the way DAs are introduced into the market plays a large role in attitude creation: companies first set expectations (through marketing and product onboarding) before consumers adopt the technology. Therefore, we inverted the hypothesised relationship.

Table 2: Grounding tasks. Task descriptions are provided in generic form due to confidentiality considerations.

Task ID	Task	Class of automation
1	Basic web search	Acquisition
2	Weather information for a specific location	Acquisition
3	Personalized information	Analysis
4	Incident-related notification	Analysis
5	Content recommendation	Decision
6	Navigation recommendation	Decision
7	Repetition of user action	Action
8	Incident-related action	Action

4 MODEL EVALUATION

4.1 Setup and data collection

To evaluate the conceptualized model, we developed an online survey. Where possible, we drew survey items from the literature cited above. We developed the remaining ones in consultation with researchers in our network who were domain experts in the relevant topics.

We validated the online data collection process and comprehensibility of the survey via two pilot surveys (N=400 and N=500) and two qualitative cognitive testing sessions (N=4 each), where participants were asked to think out loud while taking the survey and raise any questions or concerns. We then finalized the questionnaire based on the response patterns, participants' feedback, and internal consistency of the constructs.

The survey consisted of five sections: 1) screener for basic demographic questions and awareness of DAs, 2) questions about

Table 3: Profile of participants (N=2224).

	Item	N	Pct.
Gender	Female	1158	52%
	Male	1056	48%
	Not disclosed	10	<1%
Age	18-24	231	10.3%
	25-34	413	18.6%
	35-44	441	19.8%
	45-54	385	17.3%
	55-64	339	15.2%
	65-74	308	13.8%
	75 or older	107	5.0%

perceptions of DAs, 3) questions about willingness to delegate and perceived trust and value of DAs for a given task, 4) attention check, and 5) questions about perceptions of the task. By the awareness criterion, we selected those participants that were aware of at least one of the most widely used DAs: Siri (Apple), Alexa (Amazon), Cortana (Microsoft), Google Assistant, or Bixby (Samsung). Following the requirements for SEM analysis, we included several questions to measure each of the latent factors. The display order of questions in each part of the survey was randomized to ensure greater reliability of the measures [11, 36]. Overall, there were 53 Likert-type items in our instrument. Each participant was randomly assigned one of the tasks to ground their answers.

We recruited our survey respondents through the Cint panel and administered the survey through Qualtrics. Respondents were compensated according to the platform rules. For this initial evaluation of our model, we required all the participants to be based in the US. In total, 2977 responses were collected for our main survey.

4.2 Data description and participants

Responses were checked for quality and applicability for analysis. We discarded data from 143 respondents who did not consent to participate, 159 who did not satisfy the screener criteria, 258 who failed the attention check, and 193 with a completion time below 3.5 minutes (this would have required each question to be answered in 4 seconds; unlikely for a human with regular reading and clicking speeds). The final sample size comprised 2224 valid responses. Participant details are summarized in Table 3. During the model development, we split the data in 1724/500 train/test subsets to make sure that our iterations of model adjustments did not lead to overfitting the data. As the model fit was satisfactory for both subsets, further in the paper we report the results obtained by fitting the model with the whole dataset.

4.3 Analysis method

To evaluate the conceptualized model, we first validated the measurement model of the latent factors applying Confirmatory Factor Analysis (CFA). The CFA results are presented in Section 4.4. We then conducted path analysis in structural equation modelling

(SEM) to examine the hypothesized causal relationships. The findings of SEM are reported in Section 4.5. We used the R-lavaan package version 0.6.6 to conduct the analysis.¹ We employed maximum likelihood estimation with robust standard errors and a Satorra-Bentler scaled test statistic (MLM estimator) to account for the non-normality of the Likert-type data [41, 42].

4.4 Model validity and reliability

We assessed the internal consistency and reliability of our model using Cronbach's alpha. These values met the cut-off point of 0.7 [25, 39] and can be found in Table 4 in Appendix A included in the supplementary material. Additionally, we examined the convergent validity of the measurement items by composite reliability (CR) based on factor loadings from the CFA [17]. The specific values are also provided in Table 4 in the supplementary material. Composite reliability for all constructs exceeded the recommended level of 0.6 [17, 19]. In summary, our model was validated as satisfactory in terms of internal consistency and reliability. Next, we performed structural equation modelling to verify the hypotheses.

4.5 Structural Equation Modeling

The structural model with standardized path coefficients is shown in Figure 2. The model fit indices, $\chi^2 = 3529$, $p < 0.001$, $df = 827$, $CFI = 0.947$, $TLI = 0.940$, $RMSEA = 0.043$, $SRMR = 0.098$, indicated adequate fit [21, 25].

We examined the path relationships in the model by working backwards. Perceived usefulness of DAs was the strongest direct predictor of willingness to delegate ($\beta = 0.600$, $p < 0.001$). Trust in DAs to complete a task also had a direct positive effect on this factor ($\beta = 0.334$, $p < 0.001$). Meanwhile, different from what we expected, perceived ease of use of DAs did not have a direct relationship with willingness to delegate, but it did increase usefulness ($\beta = 0.928$, $p < 0.001$) and also had a direct positive effect on trust ($\beta = 0.835$, $p < 0.001$).

Considering the first layer of factors, user's attitudes towards DAs showed strong positive effects on all three value-related factors: usefulness ($\beta = 0.151$, $p < 0.001$), trust ($\beta = 0.171$, $p < 0.001$), and ease of use ($\beta = 0.738$, $p < 0.001$). Contrary to our initial hypotheses, users' familiarity with DAs turned out to influence trust ($\beta = -0.096$, $p < 0.001$) and usefulness ($\beta = -0.114$, $p < 0.001$) negatively, and did not affect ease of use.

Furthermore, we found that task riskiness increased users' need for control and transparency ($\beta = 0.188$, $p < 0.001$), which, in turn, decreased trust ($\beta = -0.145$, $p < 0.001$). We discovered a new path indicating that higher task subjectivity also increased need for control and transparency ($\beta = 0.660$, $p < 0.001$), but it turned out to have small direct positive effect on trust ($\beta = 0.106$, $p < 0.001$) and usefulness ($\beta = 0.069$, $p < 0.001$), opposite to what we anticipated. We dropped the task complexity factor from the final model because it was strongly correlated with users' self-efficacy and perceived task subjectivity. Higher levels of user self-efficacy decreased the perceived usefulness of DAs ($\beta = -0.138$, $p < 0.001$) but increased perceived ease of use ($\beta = 0.114$, $p < 0.001$). Finally, instead of influencing perceived usefulness, we discovered that task frequency had a positive influence on trust ($\beta = 0.113$, $p < 0.001$).

¹<https://lavaan.ugent.be/>

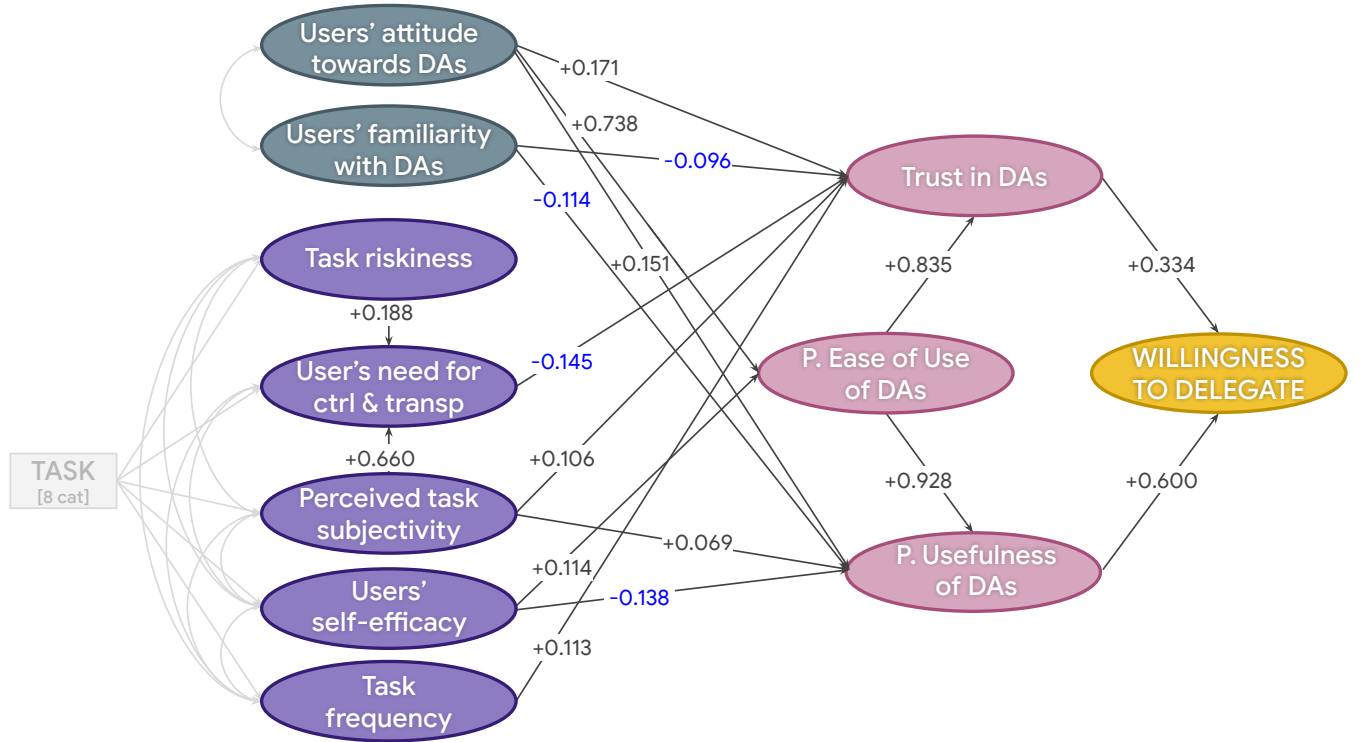


Figure 2: Structural model with standardized path coefficients. Light grey double-headed arrows show the modelled correlations between the factors. We also demonstrate how we modelled the control for tasks (dummy-coded for the model evaluation purposes).

5 DISCUSSION

The evaluation of our model largely supported our hypotheses. Below, we extract the key implications of our study and highlight opportunities for future research.

5.1 Implications

5.1.1 Mismatch between “expectations” and actual experiences. The evaluation of our model demonstrated that positive attitudes towards DAs had a large positive indirect effect on willingness to delegate, mediated through ease of use.² On the contrary, higher familiarity with DAs decreased perception of usefulness and trust in DAs, in turn reducing willingness to delegate. As attitudes approximate users’ expectations for a DA (which can be set by companies’ marketing campaigns and product setup guidance) while familiarity indicates actual use, our results reflect a mismatch between expectations and experience. This failure to meet expectations is, unfortunately, a relatively long-standing problem of this technology and an established cause of discontinued use [32]. It highlights the need for practitioners to set realistic expectations for the users, provide better feedback about system capabilities, and improve the system intelligence where it fails to deliver according to users’ expectations.

²We followed recommendations for effect size interpretation from [23]: *small* < 0.1, *medium* ≤ 0.25, *large* > 0.25.

5.1.2 Lack of control and transparency reduces trust in DAs. Lack of control and transparency reduced users’ trust in DAs, especially for subjective and risky tasks. So, even if users perceived DAs as useful for such tasks, their willingness to delegate decreased because of the loss of trust. Our findings about task riskiness are in line with previous research [31]. However, the direct positive influence of task subjectivity on trust and usefulness contradicts previous findings that users trust algorithms less for tasks that seem subjective, such as getting a movie recommendation [10]. We believe our findings might be explained by the added value of DAs for narrowing down the scope of choices that users face while completing such tasks. Further research would be beneficial to shed more light on these relationships.

5.1.3 Self-efficacy has reverse effects on usefulness and ease of use. Users perceived DAs as more useful when they struggle to complete the task without help. This is consistent with research on allocation strategies between people and automation [28]. However, in such situations, the ease of use of the DAs decreased. Practitioners should explore how they can create more intuitive interactions for scenarios when users have difficulty completing the tasks on their own, for example when they need to process an overwhelming amount of information or multitask.

5.1.4 Task frequency only influences trust. Contrary to our hypothesis, task frequency had no relationship with perceived usefulness.

This hypothesis was based on findings about the performance shaping factors in automation systems [16]. The results suggest that conclusions about automation may not always generalize to related domains such as DAs. Instead, we found that users have higher trust in DAs to complete more frequent tasks. This implies that the tasks users entrust to DAs are not the most useful, which aligns with previous studies [32, 45]. Future studies would be beneficial to investigate how trust in a given task translates to trust in using DAs more broadly.

5.2 Limitations and Future Work

This study used a survey to collect data from a US-based population. Future research should further test the model with other markets and analyse cultural differences. Moreover, applying other data sources that are not self-reported (e.g. behaviour logs) could be beneficial. Data were collected in a cross sectional survey, not longitudinally. The accuracy of the estimates relies on the causal ordering of our variables being correct. This ordering is based on theory (discussed earlier) but investigations to demonstrate this ordering are warranted.

Our study is devoted to willingness to delegate, which is a behavioural intention, but not behaviour per se. Additional analysis could be useful to investigate how our findings relate to actual use of DAs. This transition from intention to behaviour might be subject to further conditions. For example, the relationship between these two factors may be moderated by social influence and norms [33, 43], which were omitted in our work.

6 CONCLUSION

We investigated drivers of willingness to use DAs to complete various tasks. Drawing on related research on technology adoption, decision delegation, and automation, we conceptualized a model comprising key determinants of willingness to delegate, focusing on influenceable factors for practitioners. This model includes users' perceptions of different tasks, of DAs, and of their value (specifically trust, ease of use, and usefulness). Our psychometric evaluation procedure confirmed the validity and reliability of the constructs in our model and verified that the paths between them represented meaningful relationships. We found that perceived usefulness of a DA was the strongest direct predictor of willingness to delegate. Trust and positive attitudes towards DAs also increased this metric, while task riskiness, self-efficacy, and need for control and transparency had a negative impact. Critically, we found higher familiarity with digital assistants decreased users' trust in DAs and their perceived usefulness, in turn reducing willingness to delegate. This reveals a mismatch between expectations and actual experiences. Quality improvements are critical to address this gap. On top of that, practitioners should find ways to provide more control and transparency throughout the task completion process. Improving capability and usability in situations where users would struggle to complete the task without help (e.g. when their hands are busy) would help demonstrate the added value of a DA.

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